Customer Shopping Behavior Analysis

Uncovering insights from **3,900 transactions** across product categories to guide strategic business decisions.

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Dataset at a Glance

3,900 Rows

Transactions analyzed

18 Columns

Features per record

Demographics

Age, Gender, Location, Subscription

Behavior

Discounts, Frequency, Ratings, Shipping

37 missing values in the **Review Rating** column — addressed during cleaning.

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EDA with Python

01

Load & Explore

Imported with pandas; used `df.info()` and `df.describe()`

02

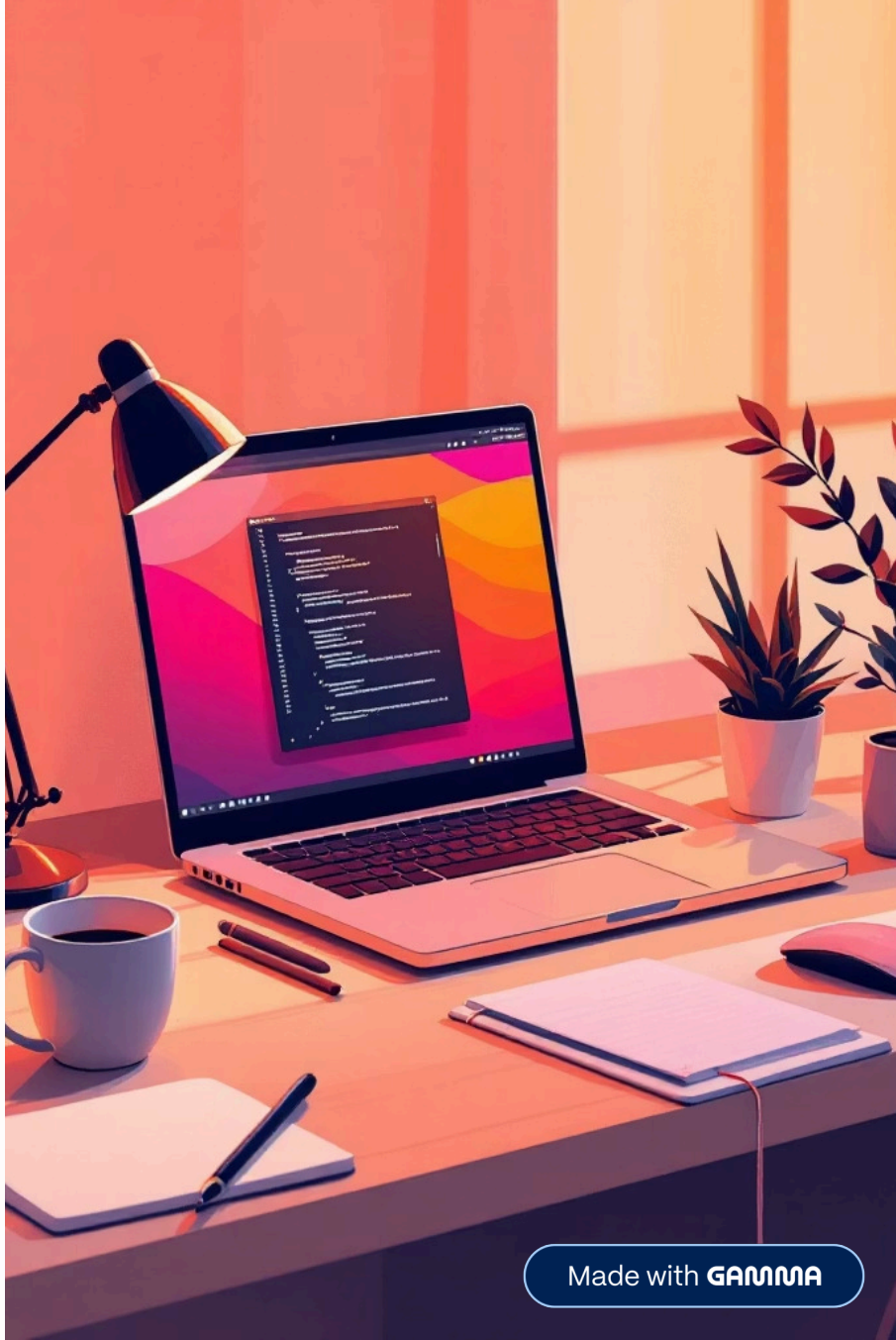
Handle Missing Data

Imputed Review Rating nulls using **median per product category**

03

Standardize Columns

Renamed to snake_case for readability



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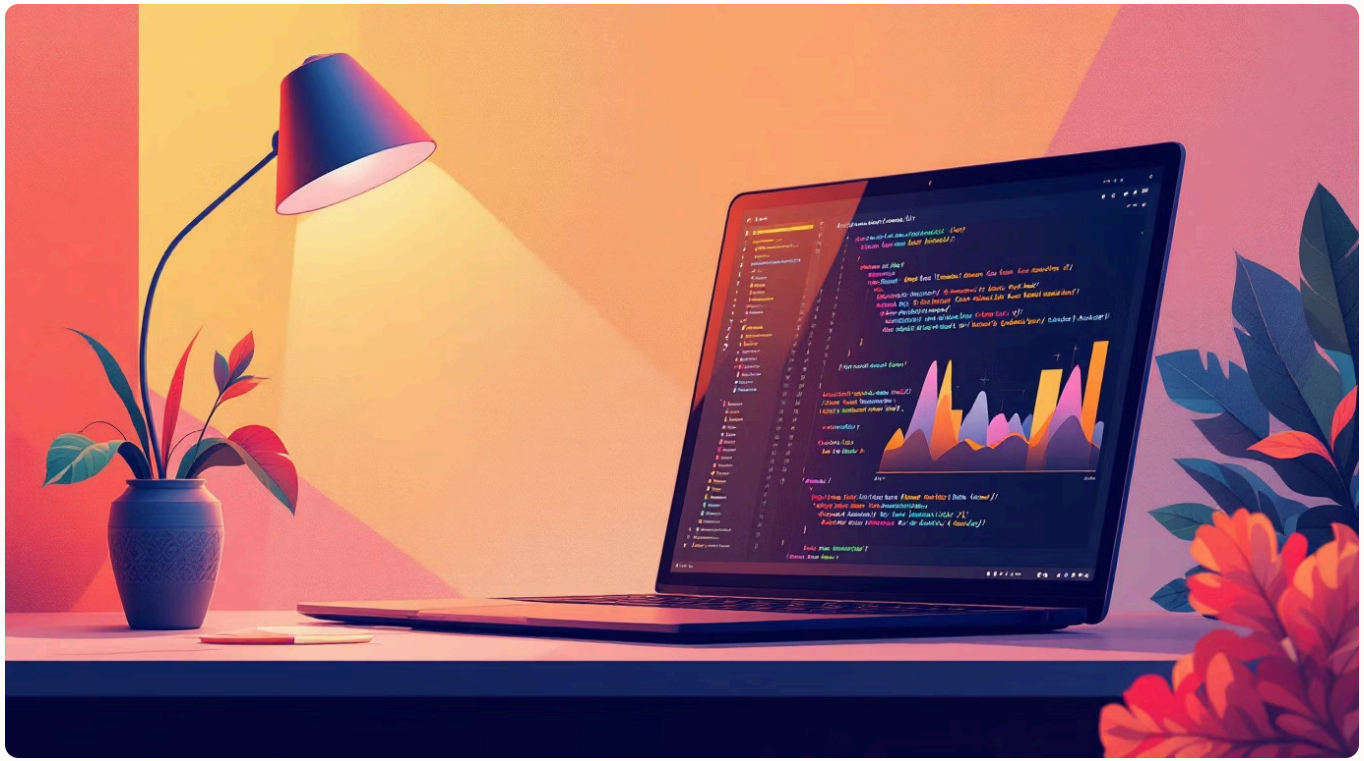
Feature Engineering



New Features Created

age_group

Customers binned into age brackets

purchase_frequency_days

Derived from purchase history data

Dropped promo_code_used

Redundant with `discount_applied`

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SQL Analysis — Key Business Questions

Revenue by Gender

Male vs. female total revenue compared

High-Spend Discount Users

Customers using discounts yet spending above average

Top 5 Products by Rating

Highest average review ratings identified

Shipping Type Comparison

Standard vs. Express average purchase amount

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More SQL Insights

Subscribers vs. Non-Subscribers

Average spend & total revenue by subscription status

Discount-Dependent Products

Top 5 products with highest % discounted purchases

Customer Segmentation

New, Returning, and Loyal — classified by purchase count

Revenue by Age Group

Total revenue contribution per age bracket

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Additional SQL Queries

Top 3 Products per Category

Most purchased items within each product category

Repeat Buyers & Subscriptions

Customers with **>5 purchases** — more likely to subscribe?

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Power BI Dashboard

Interactive visuals built to present all findings — enabling stakeholders to explore spending patterns, segments, and product performance at a glance.

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Business Recommendations

Boost Subscriptions

Promote exclusive subscriber benefits

Loyalty Programs

Reward repeat buyers → move to Loyal segment

Review Discount Policy

Balance sales boosts with margin control

Targeted Marketing

Focus on high-revenue age groups & express-shipping users

Product Positioning

Highlight top-rated & best-selling products in campaigns

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Key Takeaways

3,900 transactions

cleaned, engineered, and analyzed end-to-end

Python → SQL → Power BI

Full analytics pipeline from raw data to dashboard

Actionable insights

Subscriptions, loyalty, discounts, and targeting

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